Comparative Analysis of Classical Machine Learning Models for Face Recognition

Course: CSE445 Machine Learning — Section: CSE445.6 — Group No.: 05

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***Abstract*—This paper presents a comparative face recognition system evaluated across two datasets and two experimental chapters. Chapter 1 establishes a controlled baseline on the AT&T (ORL) dataset using PCA Eigenfaces with three classical classifiers: Support Vector Machine (SVM), K-Nearest Neighbours (KNN), and Random Forest (RF). Chapter 2 extends the study to the real-world Labeled Faces in the Wild (LFW) dataset, adding a Convolutional Neural Network (CNN). SVM achieved the highest accuracy in both settings — 96.25% on AT&T and 55.28% on LFW. The CNN failed to learn on LFW (8.13%), confirming that deep learning from scratch is infeasible at this data scale. Results show that classical PCA-based methods remain competitive on small, constrained face recognition tasks, while real-world variation causes a measurable and substantial performance drop. The full source code is available at https://github.com/aspiroo/Face-Recognition-System.**

***Index Terms*—Face Recognition, PCA, Eigenfaces, SVM, KNN, Random Forest, CNN, AT&T Dataset, LFW, Hyperparameter Tuning, Cross-Validation**

I. Introduction

Face recognition is a biometric identification technique that maps a facial image to a known identity using patterns learned from labelled examples. It has wide applications in security, authentication, and surveillance. Despite decades of research it remains challenging because the same person looks very different across lighting conditions, poses, and expressions, while two different people can appear superficially similar.

The difficulty scales with dataset complexity. A system trained on studio photographs may fail entirely when deployed under real-world conditions. Quantifying this degradation is as important as optimising accuracy on a single benchmark.

This paper addresses both goals across two chapters. Chapter 1 uses the AT&T dataset — 40 subjects photographed under controlled conditions — to build and tune three classical models using PCA Eigenfaces as the feature representation. Chapter 2 applies the same models plus a CNN to LFW, which presents genuine real-world variation in lighting, pose, and background. Comparing results across both chapters gives a direct measurement of how dataset complexity erodes model performance.

The specific contributions of this work are: (1) systematic per-model PCA component selection via sweep; (2) joint hyperparameter tuning for SVM (C and γ simultaneously); (3) a direct controlled vs. real-world comparison across four model families; and (4) an experimental confirmation of CNN minimum data requirements. All code is publicly available at https://github.com/aspiroo/Face-Recognition-System.

II. Related Work

Turk and Pentland [1] demonstrated that PCA can compress facial images into a compact set of basis vectors called Eigenfaces. Each face is represented as a weighted combination of these bases, reducing thousands of pixels to a small coefficient vector suitable for classification. This remains the standard preprocessing step for classical face recognition.

Cortes and Vapnik [4] introduced SVMs as margin-maximising classifiers. Phillips [5] later showed that SVM with an RBF kernel is one of the strongest baselines for face recognition, particularly on small datasets where its generalisation properties matter most.

Breiman [7] introduced Random Forests as ensembles of decision trees, each trained on a random feature subset. An important property for this work is that RF provides a feature importance score for each input dimension, revealing which Eigenfaces drive classification.

Huang et al. [3] introduced the LFW benchmark specifically to expose the gap between controlled-condition systems and real-world deployment. Modern methods such as FaceNet [9] achieve over 99% on LFW by training on millions of images with metric learning objectives — a scale far beyond the scope of classical approaches.

III. Datasets

A. AT&T (ORL) Dataset — Chapter 1

The AT&T dataset [2] contains 400 grayscale face images from 40 subjects with exactly 10 images each, captured under varying lighting, expressions (open/closed eyes, smiling), and slight pose variation (up to 20°). Each image is 92×112 pixels in PGM format. Fig. 1 shows one sample per subject. An 80/20 stratified split yields 320 training images and 80 test images (8 and 2 per subject). Because a three-way split would leave only 1 validation image per person — too few to be statistically useful — all hyperparameter tuning used 5-fold cross-validation on the training set.

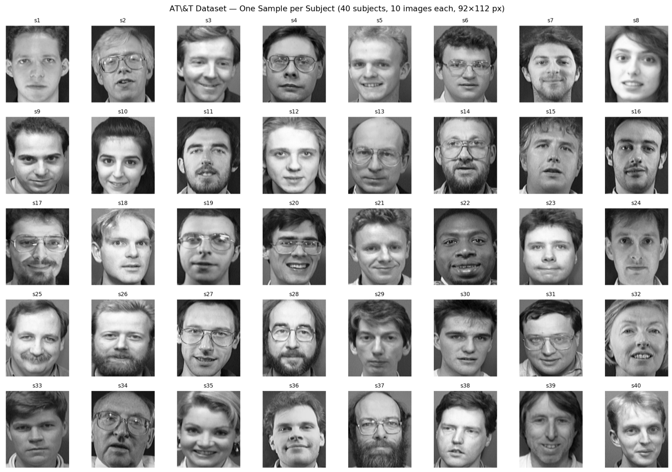

Fig. 1. One sample image per subject from the AT&T dataset (40 subjects, 10 images each). Images are 92×112 px grayscale, captured under controlled studio conditions with slight variation in lighting and expression.

B. Labeled Faces in the Wild — Chapter 2

LFW [3] contains over 13,000 face images from 5,749 individuals collected from the web under unconstrained conditions. A subset was extracted using `fetch_lfw_people` with a minimum of 40 images per subject and a cap of 100, producing 1,227 images across 19 subjects (Fig. 2). Images were resized to approximately 62×47 pixels, converted to grayscale, and normalised to $[0, 1]$. A stratified 60/20/20 split gives 736 training, 245 validation, and 246 test images — approximately 36–52 training samples per person.

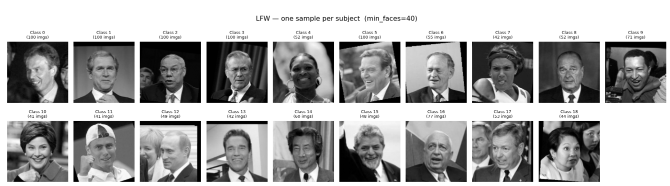

Fig. 2. One sample image per subject from the LFW subset (19 subjects). Labels show class index and image count before capping. Unlike AT&T, images vary substantially in lighting, background, pose, and image quality.

TABLE I
DATASET SUMMARY

Property	AT&T	LFW (subset)
Total images	400	1,227
Subjects	40	19
Images per person	10	41–100
Resolution	92×112	$\approx 62 \times 47$
Environment	Controlled	Real-world
Split	80/20	60/20/20
Training images	320 (8/person)	736 ($\approx$39/person)
Test images	80 (2/person)	246 ($\approx$13/person)

IV. METHODOLOGY

A. PCA Eigenfaces

Each image is flattened to a vector and normalised to $[0, 1]$. PCA then finds the directions of greatest variation across all training faces — called Eigenfaces — and projects each image onto these directions. The result is a short vector of weights that describes the face in terms of its similarity to each Eigenface. Mathematically, given a set of training faces, PCA finds a matrix $\mathbf{U}_k$ of k basis vectors such that the projection:

$$\tilde{\mathbf{x}} = \mathbf{U}_k^\top (\mathbf{x} - \boldsymbol{\mu}) \tag{1}$$

minimises the reconstruction error, where $\boldsymbol{\mu}$ is the mean training face. This reduces AT&T images from 10,304 dimensions and LFW images from 2,914 dimensions to a compact k-dimensional representation. PCA is always fit on training data only; validation and test images are projected using the already-learned basis.

Fig. 3 shows the top Eigenfaces from the AT&T training set. Early components capture global structure (lighting, face outline); later components encode finer details (eyes, mouth).

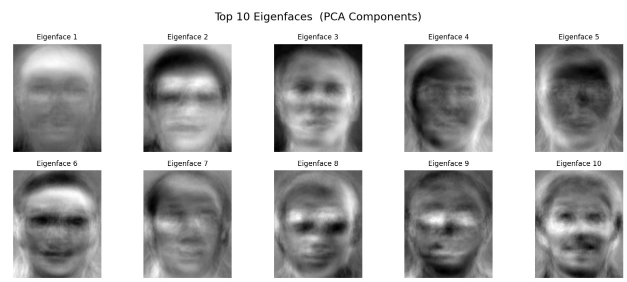

Fig. 3. Top 10 Eigenfaces from the AT&T training set. Early components (left) capture broad illumination patterns and face outline; later components (right) encode localised structure such as eye and mouth regions.

Fig. 4 shows the corresponding Eigenfaces from the LFW training set. Compared to AT&T, the basis vectors appear visibly noisier — a direct consequence of the greater diversity of lighting, backgrounds, and poses in the real-world images.

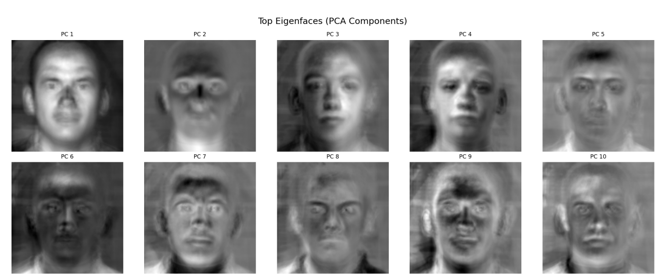

Fig. 4. Top 10 Eigenfaces from the LFW training set. The basis vectors are visibly noisier than the AT&T Eigenfaces (Fig. 3), reflecting the greater intra-class variation in lighting, pose, and background across real-world face images.

B. PCA Component Selection

Rather than fixing the number of components k arbitrarily, a sweep was run for each model independently. A fast proxy classifier evaluated 5-fold cross-validation accuracy at each candidate value of k. The value yielding the highest validation accuracy was selected for final training. Selected values: SVM (AT&T: 30, LFW: 150), KNN (AT&T: 35, LFW: 160), RF (AT&T: 70, LFW: 130). Figs. **??** through **??** show the sweep curves for each AT&T model; the LFW sweeps follow the same pattern but with higher optimal k values due to the greater intra-class variation in real-world images.

C. Support Vector Machine

SVM finds the decision boundary (hyperplane) that maximises the margin between classes [4]. For non-linearly separable data the RBF kernel is used, which measures the similarity between two points as:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\bigl(-\gamma \, \|\mathbf{x}_i - \mathbf{x}_j\|^2\bigr) \tag{2}$$

Intuitively, two faces with similar Eigenface representations (small distance) get a high similarity score, while dissimilar faces get a score near zero. The training objective balances two goals: maximise the margin between classes and minimise the number of misclassified training points. The parameter C controls this trade-off — large C tolerates fewer errors but may overfit; small C allows more errors but generalises better. The parameter γ controls how quickly similarity falls off with distance — large γ means only very similar faces influence the decision.

Both C and γ were tuned jointly via grid search, which is important because they interact: the optimal C depends on γ and vice versa. The search grid was $C \in \{0.1, 1, 5, 10, 50, 100\}$ and $\gamma \in \{\text{scale}, \text{auto}, 0.0001, 0.001, 0.005, 0.01, 0.1\}$.

D. K-Nearest Neighbours

KNN classifies a test image by finding its K most similar training images and taking a majority vote [6]. Cosine distance was used rather than Euclidean distance:

$$d(\mathbf{x}_i, \mathbf{x}_j) = 1 - \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \, \|\mathbf{x}_j\|} \tag{3}$$

Cosine distance measures the angle between two Eigenface vectors. This makes it insensitive to the overall brightness of the image — a face photographed under bright light and the same face photographed under dim light point in the same direction in PCA space, even though their magnitudes differ. The parameter K was tuned by evaluating $K \in \{1, 2, \ldots, 20\}$ via 5-fold cross-validation.

E. Random Forest

RF builds a large number of decision trees, each trained on a random sample of the training data and using only a random subset of features at each split [7]. The final prediction is made by majority vote across all trees:

$$\hat{y} = \arg\max_c \sum_{t=1}^{T} \mathbf{1}[h_t(\mathbf{x}) = c] \tag{4}$$

where $h_t(\mathbf{x})$ is the prediction of the t-th tree. Using many random trees reduces variance compared to a single tree. RF also provides a feature importance score indicating which Eigenfaces were most useful for splitting. The number of trees T and the feature subset size m were tuned via grid search.

F. Convolutional Neural Network

The CNN was trained directly on raw image patches without PCA, using three convolutional blocks (32, 64, 128 filters) each followed by ReLU activation and 2×2 max-pooling. Global Average Pooling (GAP) then averaged each feature map to a single value, producing a compact representation that was fed into a Dense softmax output layer. GAP was chosen over flattening because it reduces the number of trainable parameters by a factor of 256, which is critical when training data is limited. The network was optimised using Adam (lr $= 10^{-3}$) with early stopping on validation loss (patience = 10 epochs).

V. Results

A. Chapter 1 — AT&T

All three models performed well on the controlled AT&T dataset. Tables II and III summarise the results and best hyperparameters.

TABLE II
AT&T Dataset — Performance Metrics

Model	Acc.	Prec.	Rec.	F1
SVM	96.25%	97.08%	96.25%	96.08%
KNN	98.75%	99.17%	98.75%	98.67%
RF	92.50%	92.50%	92.50%	91.42%

TABLE III
AT&T Dataset — Hyperparameters and CV Stability

Model	Best Params	CV Mean	PCA k
SVM	$C = 5$, $\gamma = 0.01$	97.50%±1.59%	30
KNN	$K = 1$, cosine	82.24%±8.75%	35
RF	$n = 700$, feat=log2	94.06%±1.82%	70

Fig. 5 shows the SVM C–γ tuning heatmap for AT&T. The optimal cell at $C = 5$, $\gamma = 0.01$ is clearly localised, with accuracy degrading in all directions — a classic illustration of the joint sensitivity of these two parameters.

Fig. 6 shows the KNN K-curve for AT&T. The curve peaks sharply at $K = 1$ and declines monotonically, reflecting that under controlled conditions the single closest Eigenface match is almost always correct.

Fig. 7 shows the RF tuning heatmap. The optimum at $n = 700$, max_feat=log2 indicates that the most aggressive feature randomisation (fewest features per split) produces the best ensemble diversity on this small dataset.

Fig. 8 shows the PCA component importances from the AT&T RF. Classification is dominated by fewer than ten early

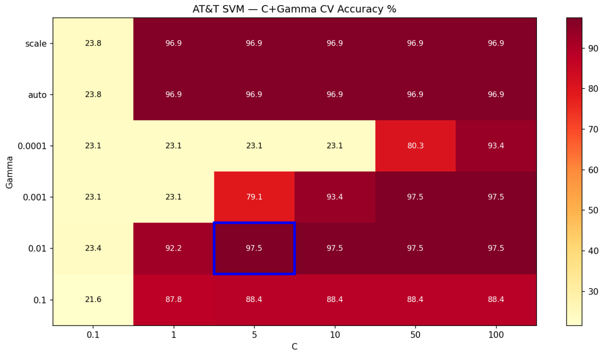

Fig. 5. AT&T SVM: Joint C–γ tuning heatmap (5-fold CV accuracy). The blue border marks the optimum at $C = 5$, $\gamma = 0.01$. Accuracy drops symmetrically away from the optimum, confirming the bias-variance interaction between C and γ.

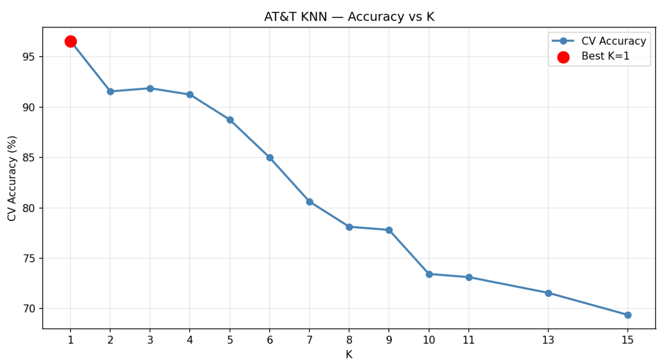

Fig. 6. AT&T KNN: CV accuracy vs. K. The peak at $K = 1$ shows that under controlled conditions the nearest Eigenface is a reliable identity signal. Larger K introduces label noise from neighbouring subjects.

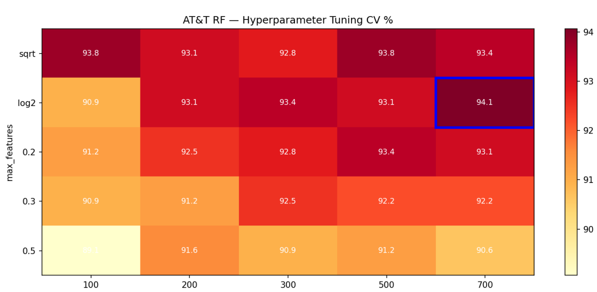

Fig. 7. AT&T RF: Hyperparameter tuning heatmap (5-fold CV accuracy). The optimum at $n = 700$ trees with log2 feature subsampling indicates that high ensemble diversity is beneficial on a small dataset.

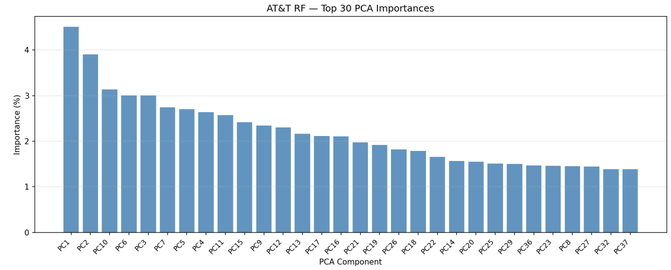

Fig. 8. AT&T RF: Top 30 PCA component importances. The distribution is highly concentrated in PC1–PC10, showing that global facial structure (captured by low-order Eigenfaces) is the primary discriminative signal.

principal components, confirming that global facial structure — not fine-grained detail — drives the decisions.

Fig. 9 shows the AT&T SVM confusion matrix. The diagonal is near-perfect, with only three off-diagonal errors across 80 test images.

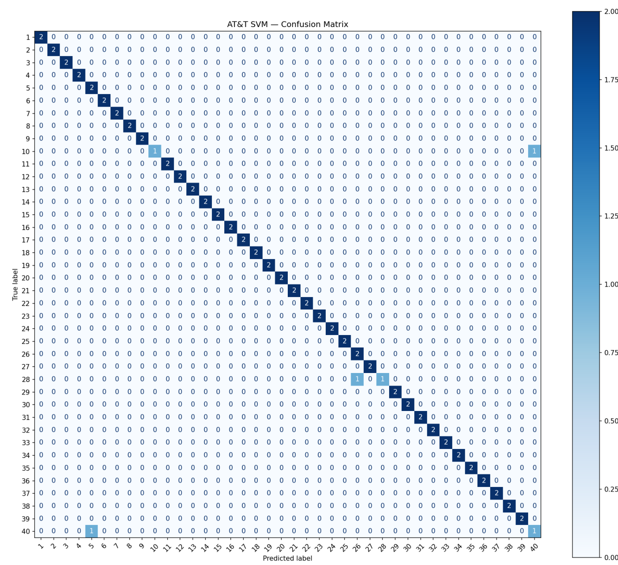

Fig. 9. AT&T SVM confusion matrix (96.25% accuracy). The near-perfect diagonal with only three errors across 80 test images demonstrates strong subject discrimination under controlled conditions.

Figs. 10 and 11 show the corresponding confusion matrices for KNN and RF. KNN's matrix is the cleanest of the three — only one error — while RF shows slightly more off-diagonal activity, consistent with its lower accuracy of 92.50%.

Fig. 12 compares accuracy, precision, recall, and F1 across all AT&T models.

B. Chapter 2 — LFW

Performance dropped substantially on LFW across all models. Tables IV and V give the full results.

Fig. 13 shows the LFW SVM tuning heatmap. The optimal region is broader than AT&T, reflecting lower hyperparameter sensitivity under real-world variation.

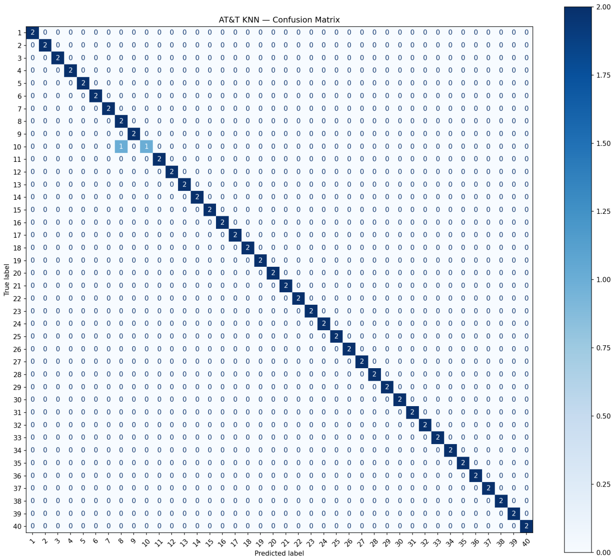

Fig. 10. AT&T KNN confusion matrix (98.75% accuracy). A single misclassification across 80 test images confirms that cosine-distance nearest-neighbour matching is nearly perfect under controlled conditions.

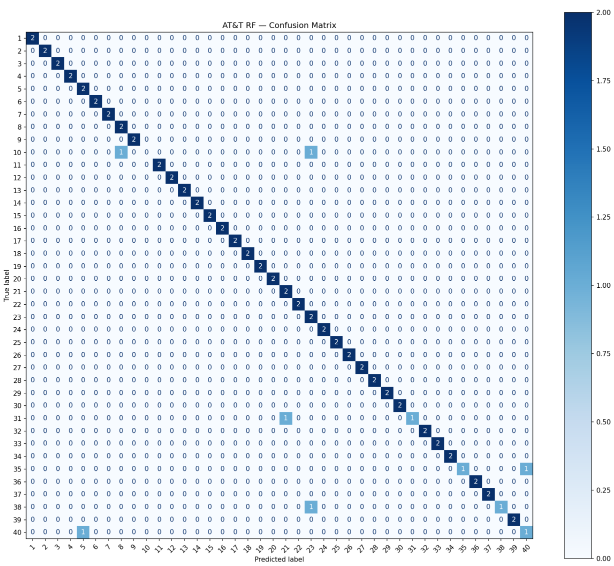

Fig. 11. AT&T RF confusion matrix (92.50% accuracy). Six errors are visible off the diagonal, more than SVM or KNN, consistent with the ensemble's higher variance on this small dataset.

TABLE IV
LFW DATASET — PERFORMANCE METRICS

Model	Acc.	Prec.	Rec.	F1
SVM	55.28%	65.53%	53.17%	56.84%
RF	40.24%	38.81%	37.12%	33.18%
KNN	39.02%	47.74%	36.59%	38.20%
CNN	8.13%	0.43%	5.26%	0.79%

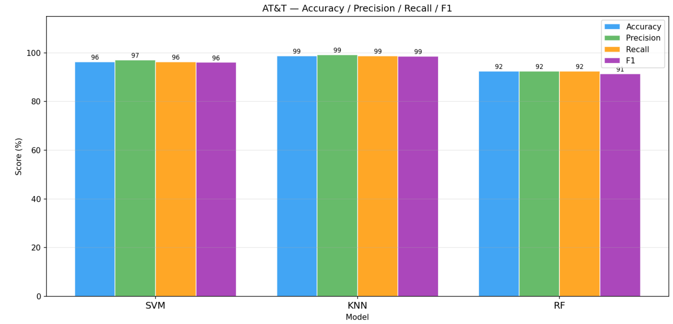

Fig. 12. AT&T model comparison: accuracy, macro precision, recall, and F1 per model. KNN achieves the highest accuracy but SVM shows the most consistent scores across all four metrics.

TABLE V
LFW DATASET — HYPERPARAMETERS AND CV STABILITY

Model	Best Params	CV Mean	PCA k
SVM	$C = 2$, $\gamma = 0.005$	57.29%±1.76%	150
RF	$n = 700$, feat=0.3	41.08%±3.86%	130
KNN	$K = 19$, cosine	37.78%±2.48%	160
CNN	3×Conv, GAP, lr=10^{-3}	N/A	N/A

Fig. 14 shows the SVM 5-fold CV scores on LFW. The low standard deviation (1.76%) confirms consistent performance across splits.

Fig. 15 shows the LFW KNN K-curve, which is the mirror image of the AT&T curve — accuracy rises steadily to a peak at $K = 19$ rather than dropping after $K = 1$.

Fig. 16 shows the LFW RF feature importances. The distribution is more spread than AT&T, with important components scattered across a wider range — consistent with LFW requiring more diverse features to discriminate 19 subjects.

Fig. 17 shows the CNN training history. Both train and validation loss decrease at the same rate and converge to the same value — the signature of underfitting rather than overfitting.

Figs. 18 through 21 show the confusion matrices for all four

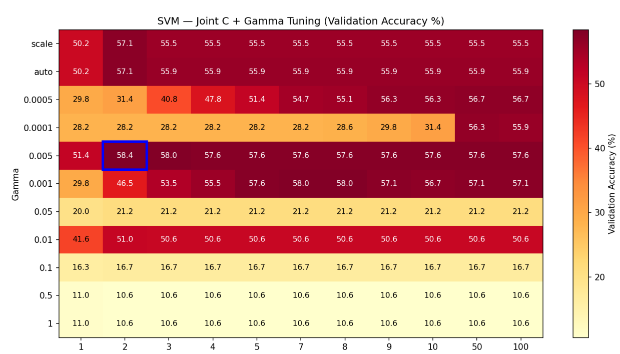

Fig. 13. LFW SVM: Joint C–γ tuning heatmap (validation set accuracy). The broader optimal region compared to AT&T indicates lower sensitivity to exact hyperparameter values when intra-class variation is high.

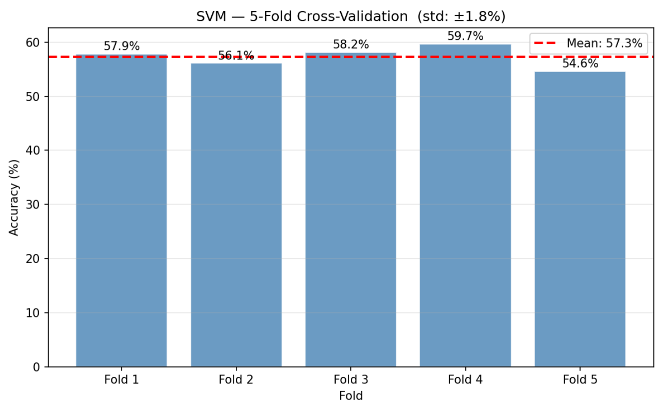

Fig. 14. LFW SVM: 5-fold CV fold accuracies with mean line. The low standard deviation (±1.76%) confirms that the SVM generalises consistently across different random splits of the training data.

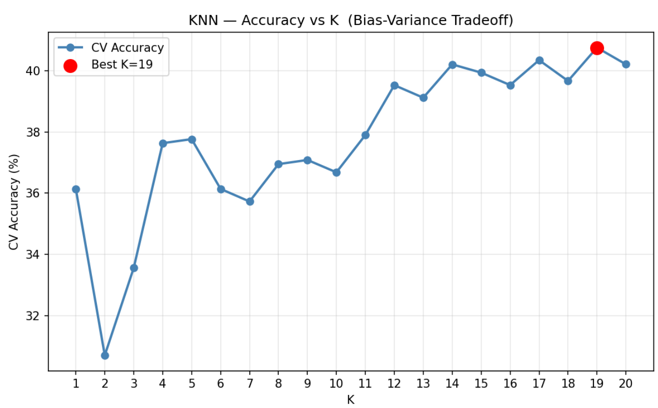

Fig. 15. LFW KNN: CV accuracy vs. K. In contrast to AT&T, the optimal $K = 19$ shows that many neighbours must be consulted to overcome noisy nearest-neighbour matches caused by real-world intra-class variation.

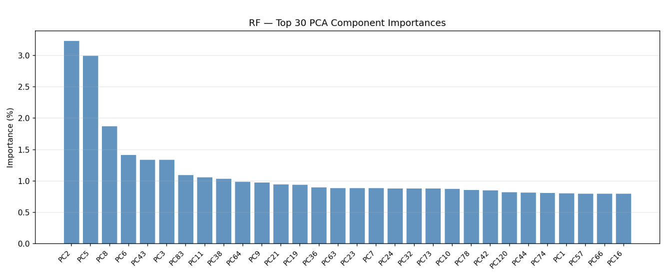

Fig. 16. LFW RF: Top 30 PCA component importances. Compared to AT&T, the important components are more spread across the spectrum, reflecting that real-world face discrimination requires a broader set of facial features.

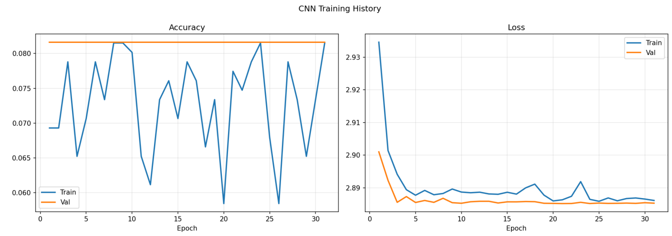

Fig. 17. LFW CNN training history. Train and validation accuracy are nearly identical throughout, confirming underfitting. The model never learned discriminative features and collapsed to predicting the most frequent class.

LFW models. The SVM matrix has a visible but imperfect diagonal; KNN and RF show more diffuse errors; the CNN matrix is almost entirely concentrated in one or two predicted columns, confirming the model predicted a single dominant class for nearly all inputs.

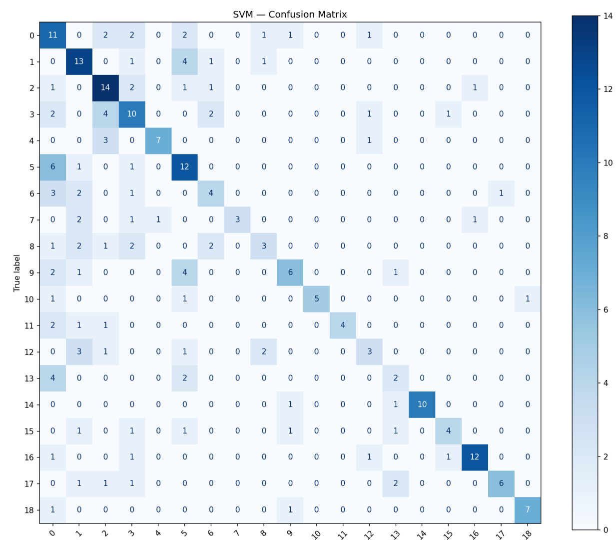

Fig. 18. LFW SVM confusion matrix (55.28% accuracy). A visible diagonal indicates correct predictions, but off-diagonal errors are more frequent than AT&T due to real-world intra-class variation. Some subjects are consistently confused with visually similar ones.

Fig. 22 shows the cross-dataset accuracy comparison, and Fig. 23 the precision/recall/F1 breakdown for LFW models.

Fig. 24 compares CV mean and test accuracy side by side for all models that reported CV scores.

VI. Discussion

A. *SVM Generalises Best Across Both Datasets*

SVM achieved the highest accuracy in both Chapter 1 (96.25%) and Chapter 2 (55.28%). The RBF kernel maps Eigenface vectors into a space where the margin between face classes can be maximised, providing strong generalisation even with limited training data. The margin depends only on a small subset of training images (support vectors), making the model robust to the high-dimensional noise inherent in PCA representations.

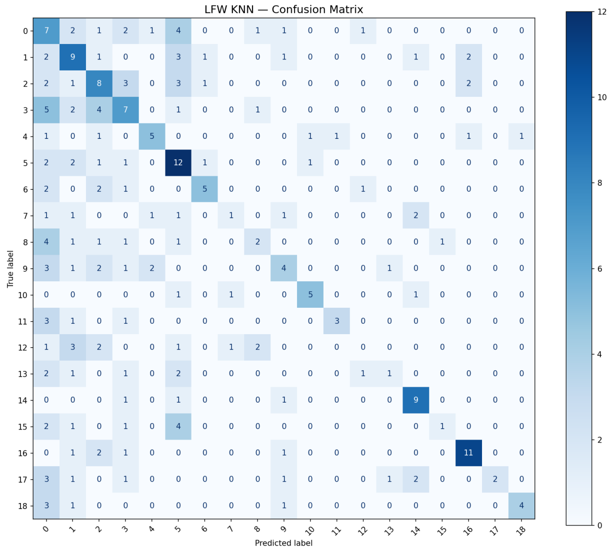

Fig. 19. LFW KNN confusion matrix (39.02% accuracy). The diagonal is weaker than SVM, with more frequent off-diagonal errors reflecting the sensitivity of nearest-neighbour matching to real-world intra-class variation across 19 subjects.

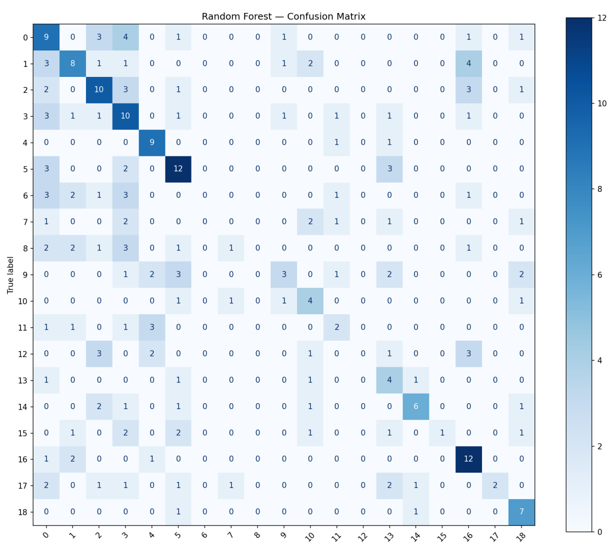

Fig. 20. LFW RF confusion matrix (40.24% accuracy). Performance is similar to KNN. Errors are spread across many subjects, with the smallest classes showing the most misclassifications.

The joint C–γ tuning was essential. On LFW the optimal parameters shifted from ($C = 5$, $\gamma = 0.01$) to ($C = 2$, $\gamma = 0.005$) — a softer margin and broader kernel, reflecting the need to tolerate more within-class variation without overfitting. Sequential tuning would likely have missed this optimum.

B. KNN: Strong Under Control, Weak Under Variation

KNN achieved the best single result on AT&T (98.75%) but was the worst classical model on LFW (39.02%). Under controlled conditions the single nearest Eigenface match is nearly always the correct person. Under real-world variation

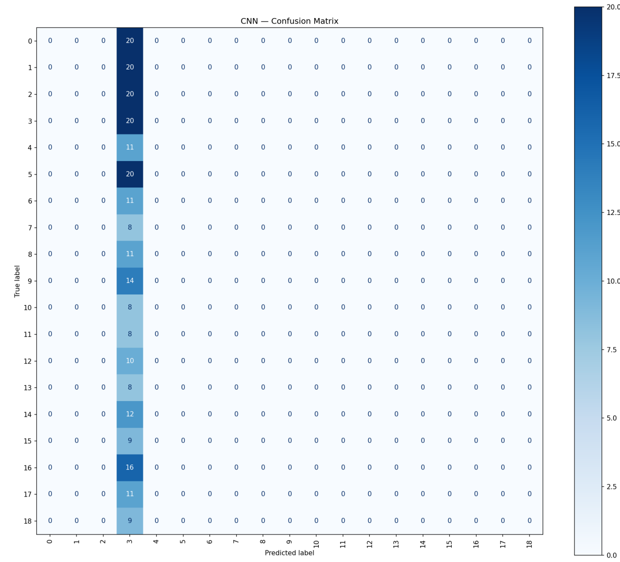

Fig. 21. LFW CNN confusion matrix (8.13% accuracy). Nearly all predictions are concentrated in one or two columns, confirming the model collapsed to predicting the most frequent class rather than learning subject-discriminative features.

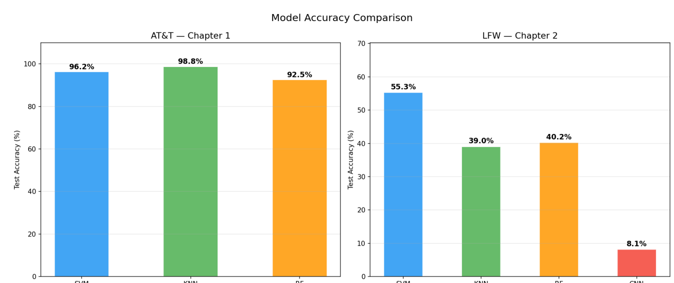

Fig. 22. Accuracy comparison across all models and both datasets. Left: AT&T (Chapter 1); Right: LFW (Chapter 2). The performance drop from controlled to real-world conditions is substantial for all models.

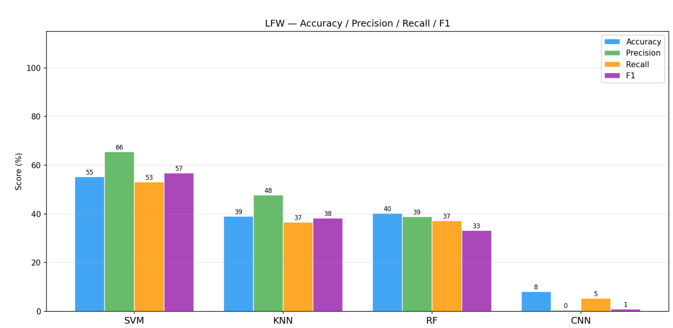

Fig. 23. LFW model comparison: accuracy, macro precision, recall, and F1. The CNN shows near-zero precision despite non-zero recall, confirming it predicted one class for nearly all inputs. SVM leads on all four metrics.

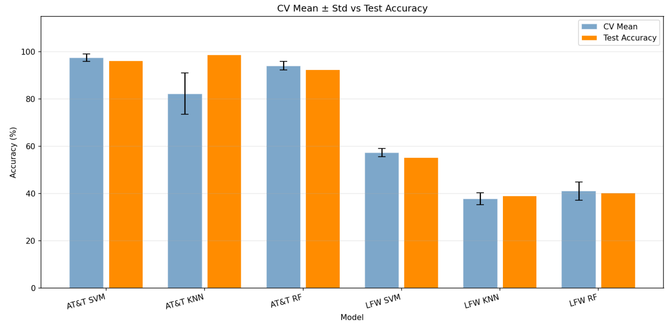

Fig. 24. CV mean $\pm$ std vs. test accuracy for all models with reported CV scores. Close agreement between CV mean and test accuracy indicates stable generalisation; the KNN AT&T gap ($K = 1$, 82% CV vs. 99% test) reflects fold-size sensitivity discussed in Section VI.

this assumption breaks down because the same person photographed differently can appear further from themselves in PCA space than from a different person in similar lighting.

The optimal K rising from 1 to 19 captures this exactly. At $K = 19$ the model consults 19 neighbours to average out noisy individual matches. The large gap between AT&T CV mean (82.24%) and test accuracy (98.75%) is explained by fold size: each CV fold has only 6–7 training images per person rather than 8, which disproportionately penalises $K = 1$.

C. Random Forest: Stable but Outperformed

RF placed third on both datasets (92.50% AT&T, 40.24% LFW). Its CV standard deviation on LFW (3.86%) was the highest of the three classical models, indicating sensitivity to training fold composition. The feature importance analysis showed classification concentrated in fewer than ten principal components on both datasets, suggesting that higher-order Eigenfaces contribute noise rather than signal at these data scales.

D. CNN Failure: Data Insufficiency

The CNN achieved 8.13% — marginally above the 5.26% random chance baseline for 19 classes — with precision of 0.43%, indicating it predicted one dominant class for nearly all inputs. Training and validation loss converged to the same value from the first epoch, confirming underfitting. A barebones experiment with no Batch Normalisation, no Dropout, and the original learning rate produced identical results, ruling out architecture as the cause.

CNNs trained from scratch typically require approximately 1,000 or more images per class to learn generalisable convolutional filters. The LFW subset provides only 40-100 training images per person — roughly two orders of magnitude short of this requirement. This is a known empirical threshold, not a failure of implementation.

E. Controlled vs. Real-World Performance Gap

The accuracy drop from AT&T to LFW was: SVM -40.97 pp, KNN -59.73 pp, RF -52.26 pp. The gap is largest for KNN, which relies most directly on the assumption that nearby images share identity — an assumption that fails under real-world illumination and pose variation. This result replicates the historical motivation for the LFW benchmark [3] and quantifies it precisely for the four model families evaluated here.

VII. Conclusion

This paper evaluated four machine learning models for face recognition across two datasets of contrasting complexity. On the controlled AT&T dataset all classical models performed well (92.50–98.75%), demonstrating that PCA-based methods are highly effective under consistent conditions. On LFW, SVM remained the strongest at 55.28% while KNN and RF plateaued near 40% and the CNN failed at 8.13%.

The principal findings are: (1) SVM with joint hyperparameter tuning generalises best across both settings; (2) KNN's performance depends critically on the assumption that nearest neighbours share identity, which fails under real-world variation; (3) RF feature importance confirms that only a handful of Eigenfaces drive face classification; (4) the CNN requires approximately two orders of magnitude more data per class than this dataset provides.

Future work should explore transfer learning with pretrained models such as FaceNet [9] or ArcFace, which address the data requirement by leveraging millions of training examples. Real-time deployment on video streams and cross-dataset generalisation are also natural extensions.

The complete implementation is available at: https://github.com/aspiroo/Face-Recognition-System.

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